

Hong, J.Y., Cho, H.J., Kim, S.T., Park, Y.S., Shin, S.H., Han, I.W., Lee, J., Heo, J.S., & Park, J.O. (2021). Comprehensive molecular profiling to predict clinical outcomes in pancreatic cancer. *Therapeutic Advances in Medical Oncology*, 13, 17588359211038478.

Abstract

Background: Pancreatic ductal adenocarcinoma (PDAC) has the worst prognosis among common cancers. The genomic landscape of PDAC is defined by four mutational pathways: *kirsten rat sarcoma virus (KRAS)*, *cellular tumor antigen p53 (TP53)*, *cyclin dependent kinase inhibitor 2A (CDKN2A)*, and *SMAD family member 4 (SMAD4)*. However, there is a paucity of data on the molecular features associated with clinical outcomes after surgery or chemotherapy.

PDAC: Pancreatic Ductal Adenocarcinoma (췌관 선암증) PDAC $\subset$ TCGA- PAAD

4 mutational pathways:

kirsten rat sarcoma virus (KRAS), cellular tumor antigen p53 (TP53), cyclin dependent kinase inhibitor 2A (CDKN2A), and SMAD family member 4 (SMAD4)

PDAC의 핵심 발암 경로는 KRAS (75%), TP53 (67%), SMAD4 (20%)로 정의되며, 이를 근거로 KRAS signaling pathway, TP53 pathway, SMAD4 pathway 3개를 생물학적 근거 기반 후보 pathway로 선정함. 췌장암 survival model 구축 시 단일 pathway만을 고려하는 것이 아니라, 복수의 pathway를 동시에 고려하는 것이 생물학적으로 정당함.

Results: *KRAS* (75%), *TP53* (67%), *CDKN2A* (12%), *SMAD4* (20%), and *ring finger protein 43 (RNF43)* (13%) were identified as significantly mutated genes. The tumor-specific

ous systematic genomic profiling of PDAC. The key mutation in PDAC is activated *KRAS*, which occurs early in carcinogenesis, whereas mutations in *SMAD4* occur later and are thought to facilitate the progression of PDAC.^{9,10} Germline and